In [1]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

In [3]:
```
df= pd.read_excel("C:\\Users\\Dell\\OneDrive\\Desktop\\STATISTICS\\Data analysis\\project\\swiggy\\swiggy_data.xlsx")
```

In [4]:
```
df.head()
```

Out[4]:

	State	City	Order Date	Restaurant Name	Location	Category	Dish Name	Price (INR)	Rating	Rating Count
0	Karnataka	Bengaluru	2025-06-29	Anand Sweets & Savouries	Rajarajeshwari Nagar	Snack	Butter Murukku-200gm	133.9	4.0	0
1	Karnataka	Bengaluru	2025-04-03	Srinidhi Sagar Deluxe	Kengeri	Recommended	Badam Milk	52.0	4.5	25
2	Karnataka	Bengaluru	2025-01-15	Srinidhi Sagar Deluxe	Kengeri	Recommended	Chow Chow Bath	117.0	4.7	48
3	Karnataka	Bengaluru	2025-04-17	Srinidhi Sagar Deluxe	Kengeri	Recommended	Kesari Bath	65.0	4.6	65
4	Karnataka	Bengaluru	2025-03-13	Srinidhi Sagar Deluxe	Kengeri	Recommended	Mix Raitha	130.0	4.0	0

In [5]:
```
df['Category'].head(20)
```

Out[5]:
```
0                  Snack
1            Recommended
2            Recommended
3            Recommended
4            Recommended
5            Recommended
6            Recommended
7            Recommended
8     North Indian Gravy
9     North Indian Gravy
10    North Indian Gravy
11    North Indian Gravy
12    North Indian Gravy
13    North Indian Gravy
14    North Indian Gravy
15    North Indian Gravy
16    North Indian Gravy
17    North Indian Gravy
18    North Indian Gravy
19    North Indian Gravy
Name: Category, dtype: str
```

In [6]:
```
df.columns
```

Out[6]:
```
Index(['State', 'City', 'Order Date', 'Restaurant Name', 'Location',
       'Category', 'Dish Name', 'Price (INR)', 'Rating', 'Rating Count'],
      dtype='str')
```

In [7]:
```
df.shape
```

Out[7]:
```
(197430, 10)
```

In [8]:
```
df.isnull().sum().sum()
```

Out[8]:
```
0
```

In [9]:
```
# total sales
total_sales = round(df['Price (INR)'].sum())
print(f"Total Sales: {round(df['Price (INR)'].sum(),2)}")
```

```
Total Sales: 53012505.77
```

In [10]:
```
# Average Rating
print(f"Average Rating: {round(df['Rating'].mean(),2)}")
```

```
Average Rating: 4.34
```

In [11]:
```
# Average order Value
print(f"Average order value: {round(total_sales/df.shape[0],2)}")
```

```
Average order value: 268.51
```

In [12]:
```
# Ratings Count
print(f"Total number of customer reviews: {df['Rating Count'].sum()}")
```

```
Total number of customer reviews: 5591574
```

In [13]:
```
# Total Orders
print(f"Total Orders: {df.shape[0]}")
```

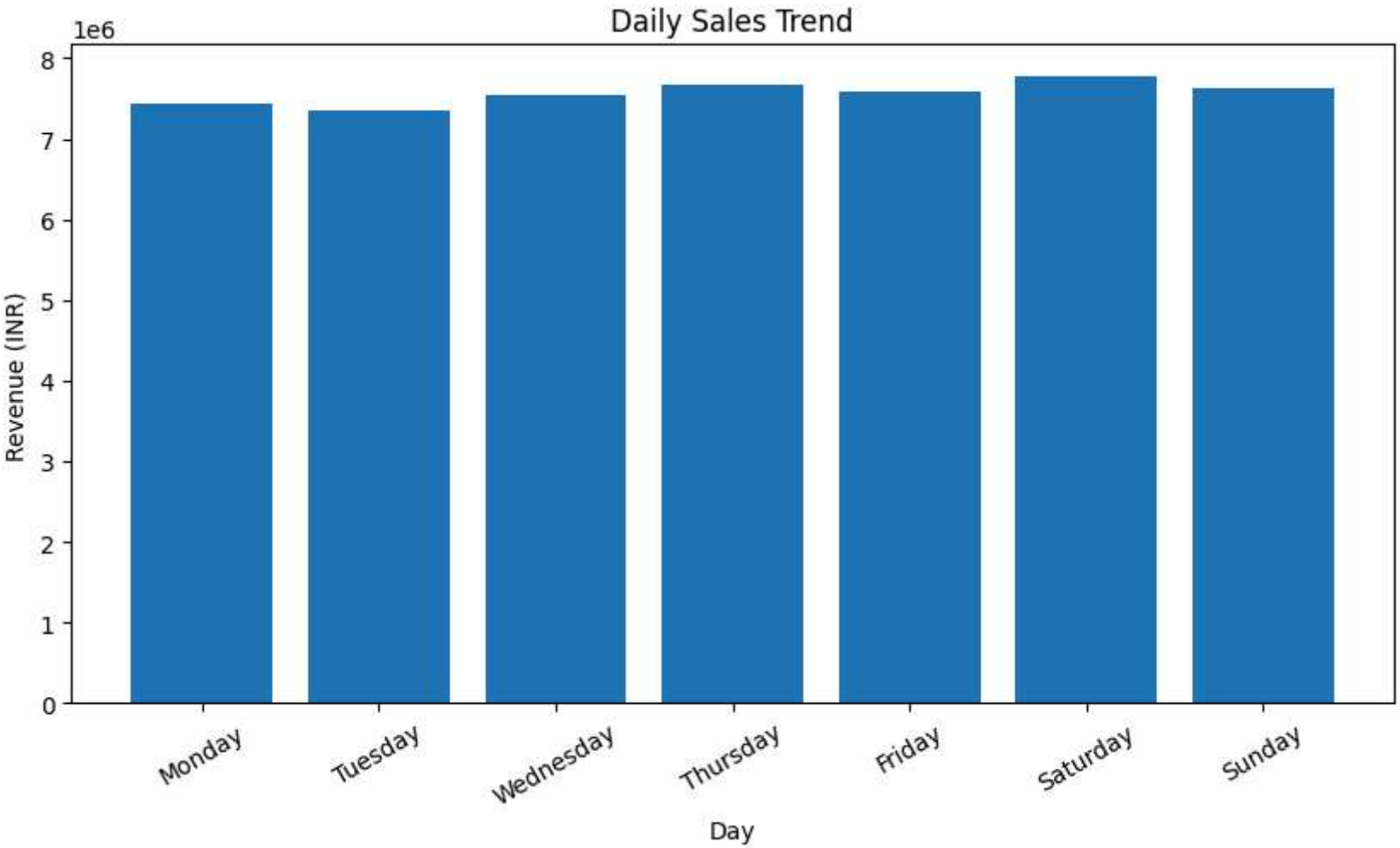
```
Total Orders: 197430
```

Charts

In [14]:
```
# Monthly Sales Trend
df['Order Date'] = pd.to_datetime(df['Order Date'])
df['YearMonth'] = df['Order Date'].dt.to_period("M").astype(str)
monthly_revenue = df.groupby('YearMonth')['Price (INR)'].sum().reset_index()

plt.figure()
plt.plot(monthly_revenue['YearMonth'], monthly_revenue['Price (INR)'])
plt.xticks(rotation=45)
plt.xlabel("Month")
plt.ylabel("Revenue (INR)")
plt.title("Monthly Sales Trend")
plt.tight_layout()
plt.show()
```

In [15]:
```
# Daily Sales Trend
df['DayName'] = pd.to_datetime(df['Order Date']).dt.day_name()

daily_revenue = (
    df.groupby('DayName')['Price (INR)'].sum().reindex(['Monday','Tuesday','Wednesday','Thursday','Friday','Saturday
)

plt.figure(figsize=(10,5))
plt.bar(daily_revenue.index, daily_revenue.values)
plt.title("Daily Sales Trend")
plt.xlabel("Day")
plt.ylabel("Revenue (INR)")
plt.xticks(rotation=30)
```

```
plt.show()
```



In [16]:
```
# Total Sales by Food Type (Veg vs Non-Veg)
non_veg_keywords = [
    "chicken", "egg", "fish", "mutton",
    "prawn", "biriyani", "kabab", "kebab",
    "non_veg", "non veg"
]
```

```
df['Food Category'] = np.where(
    df['Dish Name'].str.lower().str.contains("|".join(non_veg_keywords), na=False),
    "Non-Veg",
    "Veg"
)

food_revenue=df.groupby("Food Category")['Price (INR)'].sum().reset_index()
food_revenue
```

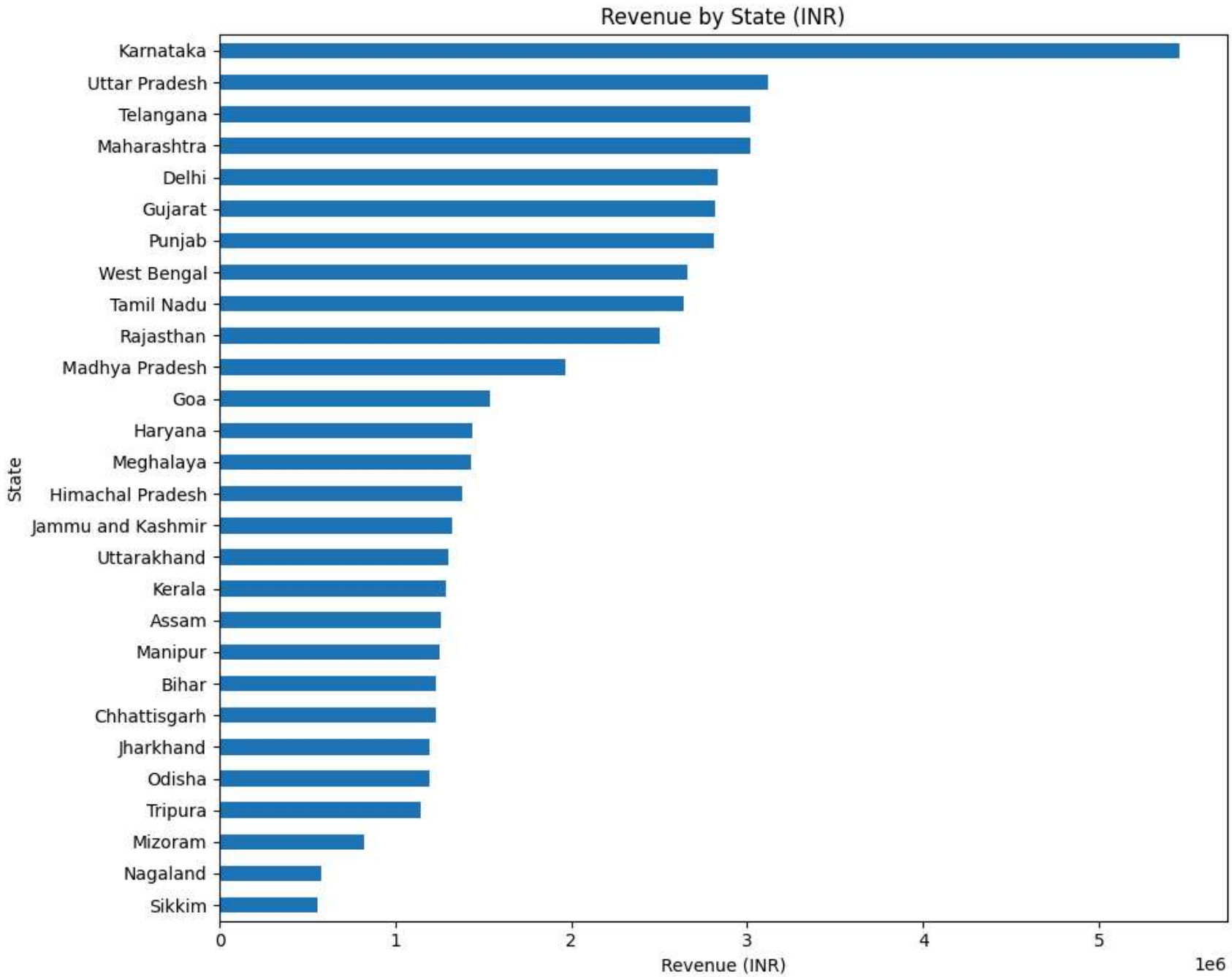
Out[16]:

	Food Category	Price (INR)
0	Non-Veg	18205776.43
1	Veg	34806729.34

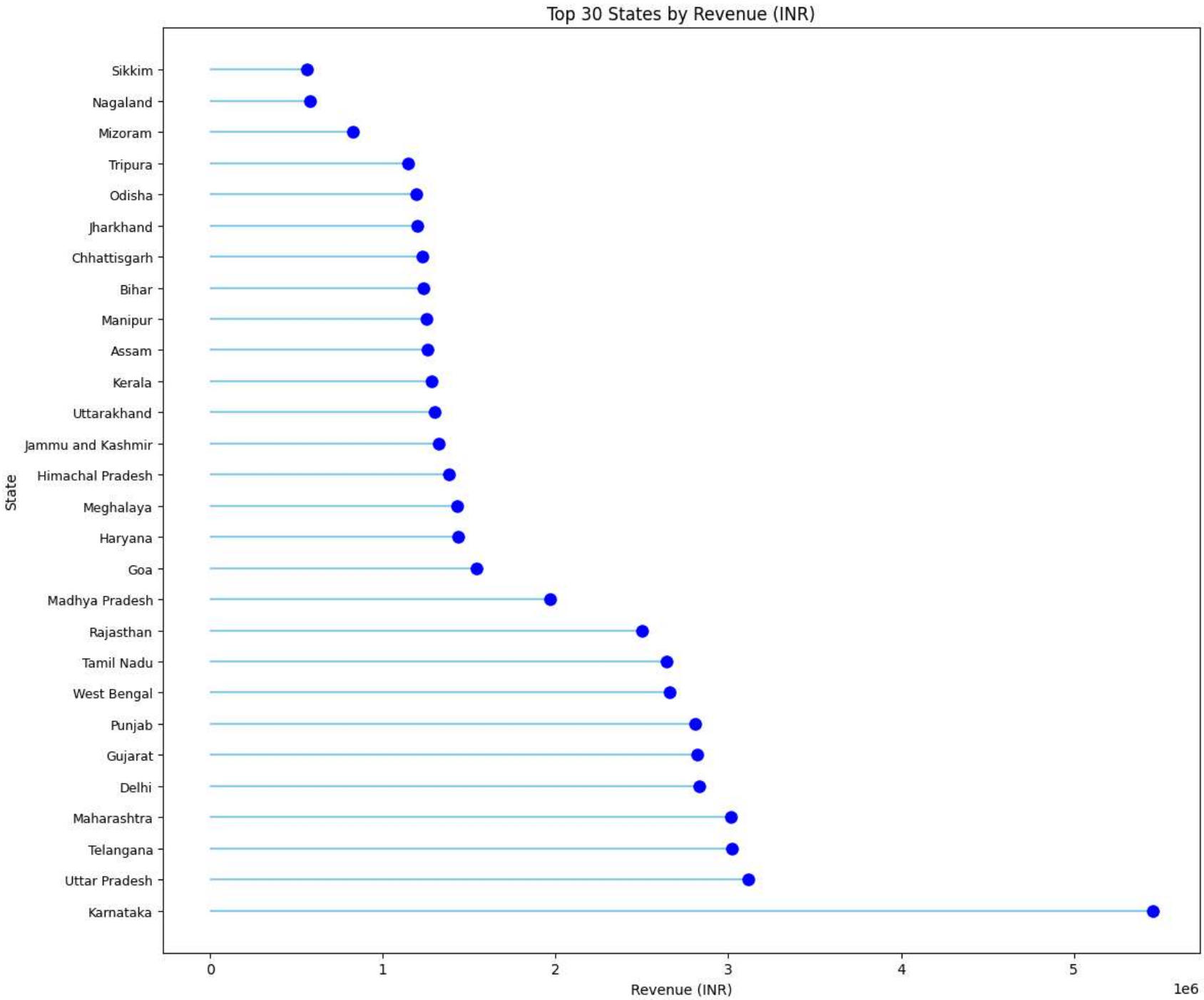
In [18]:

```
from pywaffle import Waffle

values = food_revenue['Price (INR)'].values
categories = food_revenue['Food Category'].values
colors = ['#2E7D32', '#C62828']

fig = plt.figure(
    FigureClass=Waffle,
    rows=5,
    columns=10,
    values=values,
    labels=[f"{cat}: ₹{val:,.0f} ({val/values.sum()*100:.1f}%)"
            for cat, val in zip(categories, values)],
    colors=colors,
    legend={'loc': 'upper left', 'bbox_to_anchor': (1, 1)},
    figsize=(10, 5)
)

plt.title('Revenue Contribution: Veg vs Non-Veg', fontsize=14, pad=20)
plt.tight_layout()
plt.show()
```

In [34]:
```
# Total Sales by State

state_revenue = df.groupby("State")["Price (INR)"].sum().sort_values()

state_revenue.plot(kind='barh', figsize=(10, 8), title="Revenue by State (INR)")
plt.xlabel("Revenue (INR)")
plt.ylabel("State")
plt.tight_layout()
plt.show()
```

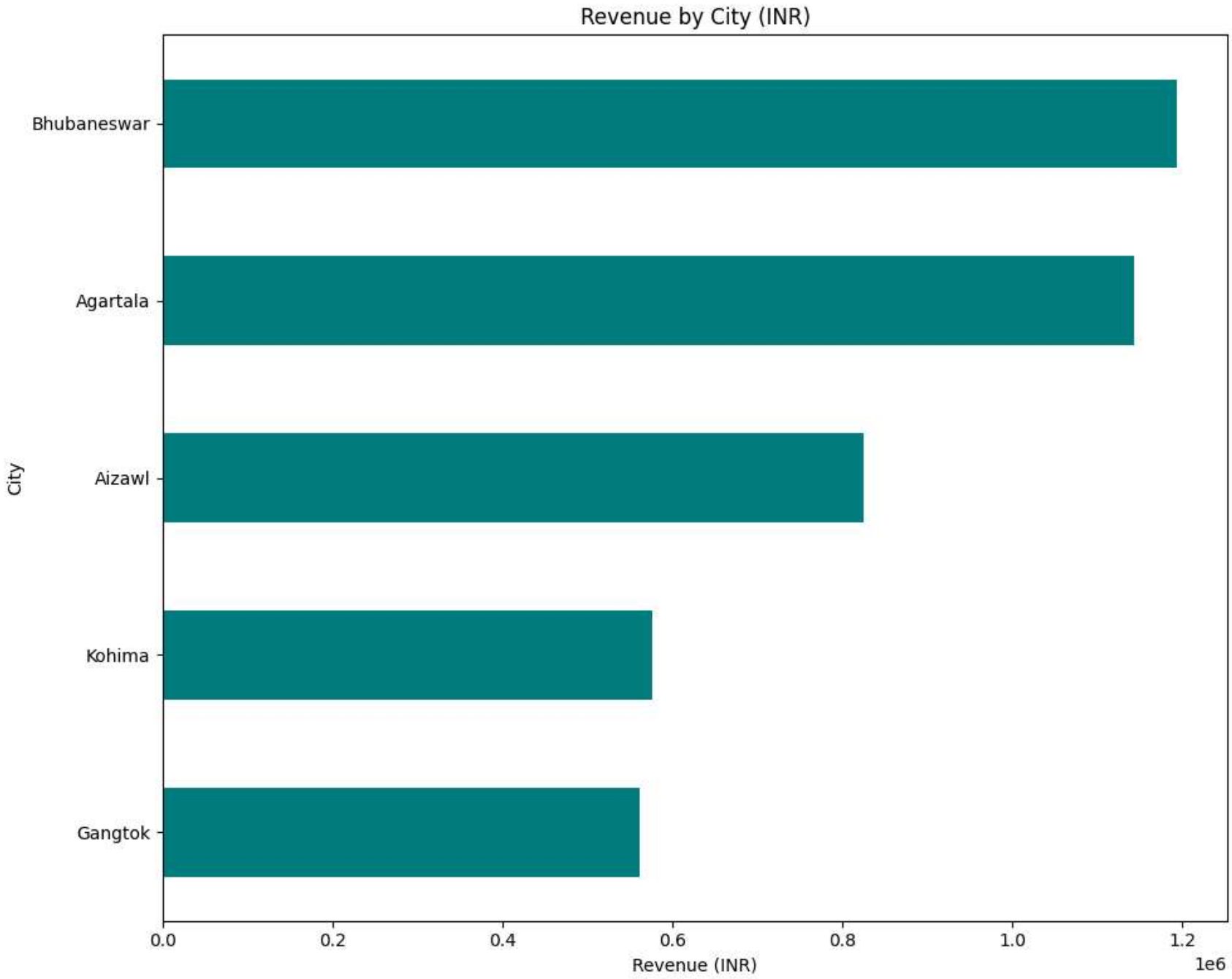

Revenue by State (INR)
Karnataka
Uttar Pradesh
Telangana
Maharashtra
Delhi
Gujarat
Punjab
West Bengal
Tamil Nadu
Rajasthan
Madhya Pradesh
Goa
Haryana
Meghalaya
Himachal Pradesh
Jammu and Kashmir
Uttarakhand
Kerala
Assam
Manipur
Bihar
Chhattisgarh
Jharkhand
Odisha
Tripura
Mizoram
Nagaland
Sikkim
State
0
1
2
3
4
5
Revenue (INR)
1e6

In [33]:
```
state_revenue = df.groupby("State")["Price (INR)"].sum().sort_values(ascending=False)

plt.figure(figsize=(12, 10))
plt.hlines(y=range(len(state_revenue)), xmin=0, xmax=state_revenue.values, color='skyblue')
plt.plot(state_revenue.values, range(len(state_revenue)), 'o', markersize=8, color='blue')
plt.yticks(range(len(state_revenue)), state_revenue.index, fontsize=9)
plt.title("Top 30 States by Revenue (INR)")
plt.xlabel("Revenue (INR)")
plt.ylabel("State")
plt.tight_layout()
plt.show()
```


Top 30 States by Revenue (INR)
Sikkim
Nagaland
Mizoram
Tripura
Odisha
Jharkhand
Chhattisgarh
Bihar
Manipur
Assam
Kerala
Uttarakhand
Jammu and Kashmir
Himachal Pradesh
Meghalaya
Haryana
Goa
Madhya Pradesh
Rajasthan
Tamil Nadu
West Bengal
Punjab
Gujarat
Delhi
Maharashtra
Telangana
Uttar Pradesh
Karnataka
State
0
1
2
3
4
5
Revenue (INR)
1e6

In [39]:

```
# Top 5 Cities by Sales
top_city = df.groupby("City")["Price (INR)"].sum().sort_values().head(5)

top_city.plot(kind='barh', figsize=(10, 8), title="Revenue by City (INR)", color='teal')
plt.xlabel("Revenue (INR)")
plt.ylabel("City")
plt.tight_layout()
plt.show()
```


Revenue by City (INR)
Bhubaneswar
Agartala
Aizawl
Kohima
Gangtok
City
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1e6
Revenue (INR)

In [ ]: